Coupled FEM/BEM: Open-Domain Conditions in 2D Electrostatics

Computational Electromagnetics

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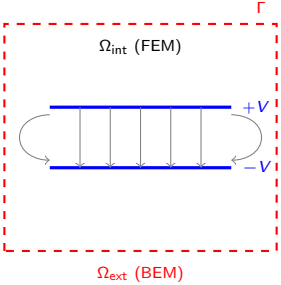
The Problem: Parallel-Plate Capacitor in 2D

Solve Laplace's equation in the whole plane:

$$\Delta u = 0 \quad \text{in } \mathbb{R}^2 \setminus \{\text{plates}\}$$

with Dirichlet conditions on the plates:

$$u = +V \text{ (top)}, \qquad u = -V \text{ (bottom)}$$



Goal: compute the electric field *both inside and outside* the capacitor, capturing the fringing effects at the plate edges.

Why Not Pure FEM?

Pure FEM approach

Truncate the domain; impose $u = 0$ on the artificial boundary Γ.

Drawbacks:

- Requires a **very large domain** to minimize the truncation error (Γ must be placed at $\sim 100\, L_y$ from the plates).
- Leads to **large meshes** and long computation times.
- Cannot represent the solution **up to infinity**: introduces an unavoidable approximation.

Key idea

Use **FEM** inside a small domain and **BEM** to handle the exterior exactly — including infinity.

FEM Weak Formulation

Starting from the Laplace problem on a bounded domain Ω:

$$\Delta u = 0 \quad \Longrightarrow \quad \int_\Omega \nabla u \cdot \nabla v \, dx - \underbrace{\int_{\partial\Omega} \frac{\partial u}{\partial n} v \, ds}_{\text{boundary term}} = 0 \qquad \forall v \in H^1(\Omega)$$

After discretisation with basis $\{v_i\}_{i=1}^{N_h}$, writing $u \approx \sum_j u_j v_j$ and $\dfrac{\partial u}{\partial n} \approx \sum_j \dfrac{\partial u_j}{\partial n} v_j$:

$$S\,\mathbf{u} \; - \; T\,\frac{\partial \mathbf{u}}{\partial n} = \mathbf{0}$$

where $S_{ij} = \displaystyle\int_\Omega \nabla v_i \cdot \nabla v_j \, dx$ and $T_{ij} = \displaystyle\int_{\partial\Omega} v_i \, v_j \, ds$.

Variable Count — The Key Issue

How many unknowns?

The system provides N_h equations. The unknowns are:

- u_j, $j = 1, \ldots, N_h$ nodal potential values
- $\partial u_j / \partial n$, $j = 1, \ldots, N_h$ normal-derivative coefficients

$$N_h \text{ equations} \quad \text{vs} \quad 2N_h \text{ unknowns} \quad \Rightarrow \quad \textbf{under-determined!}$$

Rescue: Lagrange nodal basis functions satisfy $v_i(x_j) = \delta_{ij}$, so

$$v_i\big|_{\partial\Omega} = 0 \quad \text{for every } \textit{interior} \text{ node } i \quad \Longrightarrow \quad T_{ij} = 0 \text{ if } i \text{ or } j \text{ is interior.}$$

Variable reduction

- Normal-derivative unknowns at *interior* nodes drop out of the system.
- System: N_h equations, $N_h + N_h^{\partial}$ unknowns. Here BEM provide N_h^{∂} equations.

FEM Block Structure

Partitioning nodes into **interior** (I) and **boundary** (B):

$$\underbrace{\begin{bmatrix} S_{II} & S_{IB} \\ S_{BI} & S_{BB} \end{bmatrix}}_{\text{FEM stiffness}} \begin{bmatrix} u_I \\ u_B \end{bmatrix} - \begin{bmatrix} \mathbf{0} \\ T_{BB} \dfrac{\partial u_B}{\partial n} \end{bmatrix} = \mathbf{0}$$

- Row I: closed — $\partial u / \partial n$ does not appear.
- Row B: **open** — involves the unknown $\partial u_B / \partial n$ (will be supplied by BEM).

BEM — Boundary Integral Equation

Key idea: reformulate the Laplace problem on $\partial\Omega$ only, using the fundamental solution $G(x,y) = \frac{1}{2\pi}\log|x-y|$ in 2D.

Green's third identity $\Rightarrow$ **Boundary Integral Equation** (BIE):

$$\frac{1}{2}\,u(y) = \underbrace{\int_{\partial\Omega} G(x,y)\,\frac{\partial u}{\partial n}(x)\,ds_x}_{\mathcal{V}(\partial u/\partial n)(y)} - \underbrace{\int_{\partial\Omega} \frac{\partial G}{\partial n_x}(x,y)\,u(x)\,ds_x}_{\mathcal{K}(u)(y)} \qquad \forall\, y \in \partial\Omega$$

After collocation discretizations with boundary basis $\{\psi_j\}$:

$$H\,u_B = G\,\frac{\partial u_B}{\partial n} \qquad H = \tfrac{1}{2}M + K, \quad G \text{ symmetric}$$

BEM provides **exactly** N_h^∂ equations relating u_B and $\partial u_B/\partial n$ — the missing relations from FEM.

Coupled System

Replace the open FEM boundary rows with the BEM equations:

$$\begin{bmatrix} S_{II} & S_{IB} & 0 \\ S_{BI} & S_{BB} & -T_{BB} \\ 0 & H & -G \end{bmatrix} \begin{bmatrix} u_I \\ u_B \\ \partial u_B / \partial n \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

- **Row 1 (FEM, interior):** governs u inside Ω.
- **Row 2 (FEM, boundary):** couples u_I, u_B with the flux.
- **Row 3 (BEM):** closes the system via integral operators on $\partial\Omega$.

Elimination and Reduced System

Assume G invertible:

$$\frac{\partial u_B}{\partial n} = -G^{-1} H\, u_B$$

Substitute into the FEM boundary row $\Rightarrow$ **reduced system of size** N_h:

$$\begin{bmatrix} S_{II} & S_{IB} \\ S_{BI} \quad S_{BB} + & \underbrace{T_{BB}\, G^{-1} H}_{\text{BEM correction (dense)}} \end{bmatrix} \begin{bmatrix} u_I \\ u_B \end{bmatrix} = - \begin{bmatrix} S_{ID} \\ S_{BD} \end{bmatrix} u_D$$

- Same size as a standard FEM system ($N_h \times N_h$).
- The BEM term $T_{BB} G^{-1} H$ is **dense** but restricted to boundary DOFs.
- In MATLAB: $G^{-1} H$ computed via the backslash operator.
- After solving: recover $\partial u_B / \partial n$ and the exterior potential via the BEM representation formula.

Setup

Geometry:

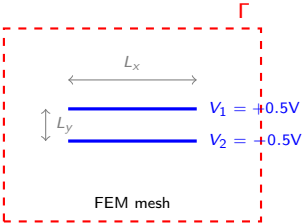
- $L_x = 0.10$ m, $L_y = 0.10$ m ($L_x/L_y = 1$)
- $V_1 = 0.5$ V (top), $V_2 = -0.5$ V (bottom)
- Γ placed at $10\,L_y$ margin on all sides
- $\varepsilon_r = 1$

Meshes:

- Coarse: $\delta = 0.005$ m ≈ 500 nodes
- Fine: $\delta = 0.002$ m ≈ 2900 nodes, 234 nodes on Γ

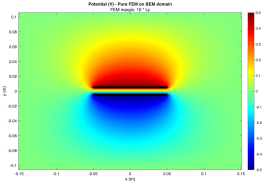
Reference: The reference is a pure FEM method with a Dirichlet condition imposed on Γ placed at $100L_y$

$$C_{\text{pure_FEM}} = ???\ \text{pF/m}$$

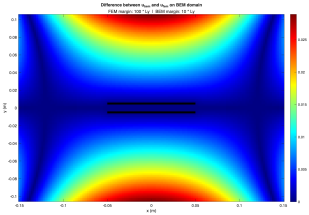


Mesh Rapresentation

FEM-BEM vs. Pure FEM — Accuracy



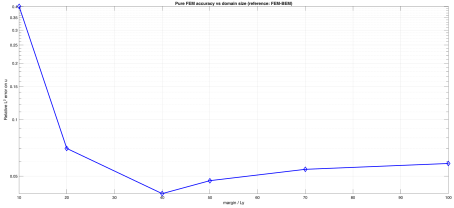
Pure FEM (100 L_y domain) — reference solution



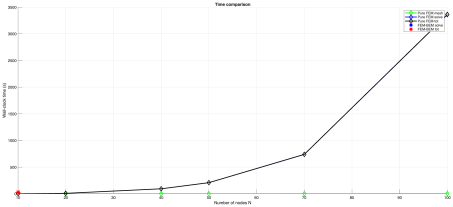
Pointwise difference: FEM vs. FEM-BEM on common domain

$$\varepsilon_{L^2} = \sqrt{\frac{\sum_{i\in\mathcal{N}_c}(u_i^{\text{FEM}} - u_i^{\text{FEM-BEM}})^2}{\sum_{i\in\mathcal{N}_c}(u_i^{\text{FEM}})^2}} \times 100\% \;=\; \mathbf{5.84}\%$$

FEM-BEM vs. Pure FEM — Accuracy & Speed



L^2 error vs. FEM domain size

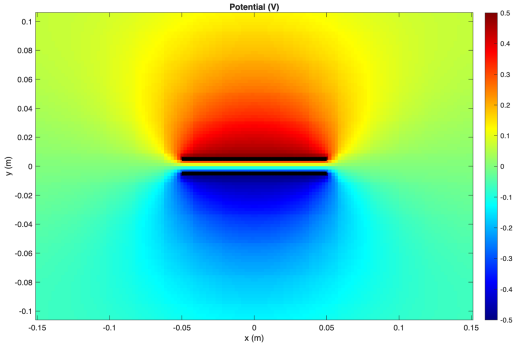


Computational time comparison

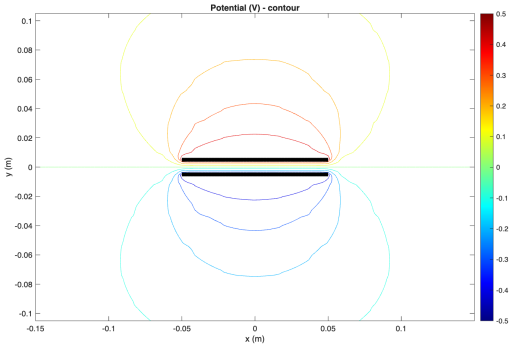
Efficiency gain *on my computer*

- FEM-BEM ($10\,L_y$ domain): **27.6 s**
- Pure FEM ($100\,L_y$ domain): **3363 s** ($\approx$ 56 min)

Electric Potential Distribution FEM-BEM



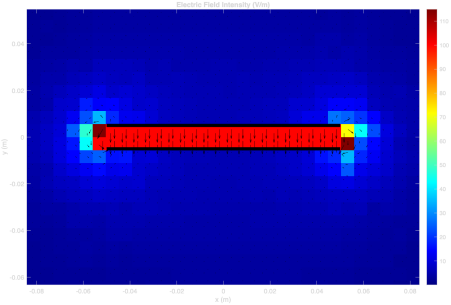
Potential map ($\delta = 0.002$ m)



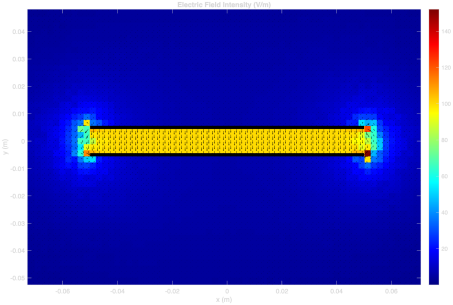
Equipotential lines

- Uniform field between plates; smooth fringing at edges.

Electric Field FEM-BEM



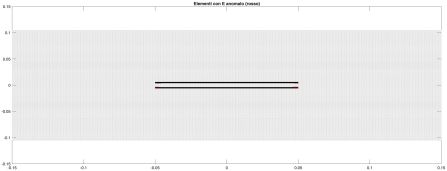
$\delta = 0.005$ m (coarse)



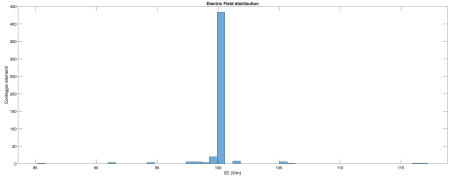
$\delta = 0.002$ m (fine)

- Central field $\approx (V_1 - V_2)/L_y = 100$ V/m as expected.
- Fine mesh resolves singularity at plate corners (physically expected).
- Coarse mesh smears the edge singularity.

Electric Field — Element Analysis (FEM-BEM)



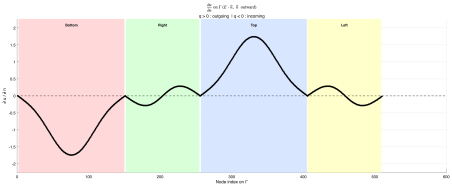
Elements with $|\mathbf{E}| > 105$ V/m



Distribution of $|\mathbf{E}|$ per element

- Anomalous elements localised at plate **corners** only.
- This is a well-known physical singularity of infinitely thin conductors — **not** a numerical artefact.
- The majority of elements have $|\mathbf{E}| \approx 100$ V/m.

Flux Through Γ (FEM-BEM)



$\partial u/\partial n$ on Γ

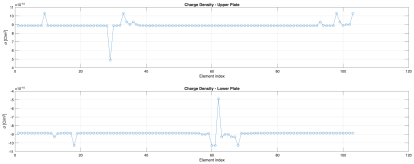
Physical expectation:

- Net flux must vanish (equal and opposite plate charges).

$$Q_{tot} = \oint_\Gamma \frac{\partial u}{\partial n}\, ds \approx -5.47 \times 10^{-13} \approx 0$$

Numerically consistent with zero — coupling condition verified.

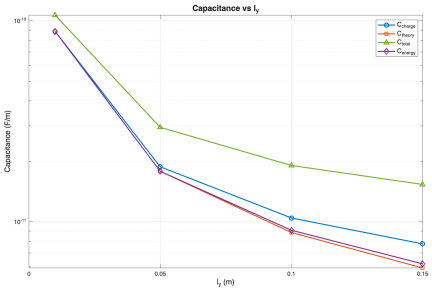
Surface Charge Density and Capacitance



Surface charge density on the plates

... come calcolare capacità, solo con carica ...

Capacitance different Ly, fixed Lx



Capacitance with different Ly

- **Method comparison:** internal energy method inside the capacitor shows excellent agreement; charge-based method slightly overestimates
- **Full domain energy:** increasing deviation from theory due to energy stored outside the capacitor

Conclusions

- A **FEM-BEM coupled solver** for 2D electrostatics has been implemented in MATLAB, targeting the parallel-plate capacitor problem.
- The coupling exploits:
  1. FEM's accuracy in the bounded interior region.
  2. BEM's ability to represent the exterior field **up to infinity**.
- Validation confirms:
  - Correct potential and field distributions.
  - Capacitance within 0.69% of the reference FEM solution.
  - Gauss's law satisfied numerically.
- **100**$\times$ **speed-up** over a comparable pure FEM simulation.